Using R to Track Flights and Weather Across Salt Lake City's Flight Network

Aleksey Kulyukin*
with
Cameron Swapp and Jürgen Symanzik

*Department of Mathematics and Statistics
Utah State University
Email: a02291810@usu.edu

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Overview

- Earlier research on Salt Lake City International Airport (SLC) and its flight network
- Flight records from the Bureau of Transportation Statistics (BTS) and and weather observations from the National Oceanic and Atmospheric Administration (NOAA)
- R-based flight processing and weather linkage
- Live demonstrations of three Shiny applications
- Denver-bound diversions to SLC and future work

Presentation repository:
https://github.com/AlKulAK/rgovys_talk_2026

Shiny applications website:
https://alkulak.github.io/rgovys_talk_2026/

Previous Research

- Kulyukin et al. (2026) analyzed domestic flights involving SLC from October 1987 through February 2026, examining airport histories, flight-network changes, and daily flight volumes.
- Three companion Shiny applications explored the flight network, flight volumes, and spatial patterns of Global Historical Climatology Network–Daily (GHCNd) weather stations. We demonstrate updated successors; the weather explorer now shows reporting availability in Global Historical Climatology Network–Hourly (GHCNh) data.
- The study identified 19 flights bound for Denver International Airport (DEN) that diverted to SLC on June 21, 2023. We revisit the case using flight movements and hourly weather observations.

BTS Reporting Carrier On-Time Performance Data

- The Bureau of Transportation Statistics (BTS) publishes monthly flight-level records for scheduled domestic passenger operations by all major carriers, beginning in October 1987.
- Records identify carriers and airports and include scheduled and actual departure/arrival times, delays, cancellations, and diversions.
- We distinguish *main flights*, with SLC as the scheduled origin or destination, from *diverted flights*, scheduled between other airports but diverted to SLC.
- Reporting requirements evolved: tail numbers and wheels-off/on times (1995), delay and cancellation causes (2003), detailed diversion records (2008), and broader carrier/airport coverage with separate marketed-codeshare reporting (2018).

NOAA GHCNh Weather Data

- NOAA's GHCNh combines observations from multiple surface weather networks.
- Reports are timestamped to the minute in Coordinated Universal Time (UTC). Reporting is not necessarily hourly: there may be gaps or multiple observations within an hour.
- Aviation reports include *METAR* (Meteorological Aerodrome Report, routine) and *SPECI* (Aviation Selected Special Weather Report, issued when specified conditions change).
- GHCNh contains 52 meteorological measurement variables, including wind, temperature, visibility, precipitation, clouds, and present weather. The separate `REM` field retains original coded reports and remarks.
- Our earlier *CHANCE* work used daily GHCNd data. GHCNh lets us examine weather reports alongside individual flight events.

`slcflights` and Flight Processing

- The `slcflights` R package downloads monthly BTS flight records and builds a local SLC-focused database, with flight data stored in annual Parquet files.
- Main and diverted flights are stored separately. Package functions can read selected years or query Parquet data lazily using Arrow.
- Separate R research scripts reconstruct flight-event sequences, including cancellations, returns to origin, and flights with multiple diversion stops.
- Scheduled and recorded event times are resolved to UTC when supported by BTS fields and airport time zones. Missing, ambiguous, or conflicting times remain flagged.

`slcflights` on GitHub:
https://github.com/AlKulAK/slcflights

Weather Processing and Flight–Weather Linkage

- R scripts identify historically applicable GHCNh stations for airports in SLC's flight network.
- Station-year observations are downloaded and processed, retaining UTC timestamps, weather measurements, source information, and original report remarks.
- Reconstructed flight departures, arrivals, and diversions are linked to weather reports using event times and historical airport–station assignments.
- Goodman and Small Griswold (2019) examined delays and cancellations using METAR reports, including at SLC. We use linked BTS and GHCNh records to reconstruct the June 2023 DEN diversion sequence.
- These scripts form the foundation for `slcweather`, a developing companion R package for `slcflights`.

1. SLC Flight Network Explorer: Overview

- Interactive map of airports and airlines connected to SLC, with flight counts for the selected network.
- Time controls select *Year* or *Month*. *All* coverage uses the full flight history; *Single* restricts the map to one year or month.
- Airline filters and optional route lines refine the map.
- Clicking an airport reveals recorded historical identities, service periods, and flight information.
- Time-series plots show counts of connected airports and airlines.
- The opening view uses *Year*, *All*, and *All Airlines*, with route lines turned off.

1. SLC Flight Network Explorer: Demo Preview

- Set *Coverage* to *Single*, select 1988, and click DEN to examine its historical Stapleton International identity.
- Move to 2025 and click DEN again to examine its contemporary Denver International identity.
- Uncheck *All Airlines*, select Southwest Airlines, and enable *Show route lines*.
- Scroll to the connected-airports chart; click 2025 to inspect the corresponding airport list.

1. SLC Flight Network Explorer: Live Demo

Webinar's companion website:

https://alkulak.github.io/rgovys_talk_2026/

2. SLC Flight Volume Explorer: Overview

- Separate views for *Main Flights* (scheduled SLC origin or destination) and *Diverted Flights* (SLC reached by diversion).
- Each point represents a date in a weekday or holiday category; hover text reports flight counts and percentages.
- The 100% reference is the same-year average daily count. For diversions, the average uses dates with recorded activity.
- The *Outlier threshold* slider highlights unusually high diversion counts without changing the plotted values.
- The app reveals daily patterns; individual flight itineraries are examined separately using the underlying records.

2. SLC Flight Volume Explorer: Demo Preview

- In *Main Flights*, hover over several Thanksgiving Days to inspect their counts and same-year percentages.
- Switch to *Diverted Flights*; identify the 100% reference and the default 750% outlier threshold.
- Move the threshold to 500%, then 1000%, and restore 750%.
- Hover over June 21, 2023, in the Wednesday facet to inspect the diverted flight count.

2. SLC Flight Volume Explorer: Live Demo

Webinar's companion website:

https://alkulak.github.io/rgovys_talk_2026/

3. SLC Airport & Weather Station Explorer: Overview

- Uses historical airport–station relationships and GHCNh reporting data for airports in the SLC flight network.
- Time controls: *Year* or *Month*, with *All* or *Single* coverage. Airports can be selected individually or in groups.
- Heatmaps reveal reporting availability for 52 GHCNh measurement-element families.
- Under *All*, heatmaps distinguish periods with no SLC service, no reported values, and reported values. Under *Single*, they show the share of applicable days with reported values.
- Clicking a heatmap cell with reported values opens reporting details; the month-origin path reaches daily and hourly counts.
- The app highlights when observations are available; actual GHCNh measurement values are examined separately.

3. SLC Airport & Weather Station Explorer: Demo Preview

- Start with *Year* and *All*; uncheck *All Airports*, select DEN, and inspect the details of a corresponding weather station.
- Scroll to DEN's annual GHCNh reporting-availability heatmap.
- Change to *Month* and *Single*; select June 2023.
- Click the `dew_point_temperature` reporting cell, then select June 21, 2023.
- Examine hourly reporting counts with airport-local and UTC time labels; then close the details drawer.

3. SLC Airport & Weather Station Explorer: Live Demo

Webinar's companion website:

https://alkulak.github.io/rgovys_talk_2026/

Behind the June 21, 2023, Diversion Spike

- The Flight Volume Explorer shows that **19** flights diverted to SLC on June 21, 2023.
- The underlying BTS records show that all 19 were bound for Denver International Airport (DEN), departing from **14 distinct origin airports**.
- Our R scripts reconstructed 149 scheduled and actual flight events and aligned them with 912 GHCNh observations across 16 airports. The complete chronology and generating code are available in the presentation repository.
- UA 557 (SJC–DEN) diverted earlier in the day and reached DEN before the evening thunderstorms. The remaining **18 flights** arrived at SLC in a late-night wave extending into June 22.

Thunderstorm Reports at Denver

All times are Mountain Daylight Time (MDT).

- **18:51:** A KDEN special weather report records thunderstorms nearby, distant lightning, and wind gusts reaching 36 knots.
- **19:22:** Another special report records thunderstorms at the airport.
- **20:41:** UA 1216 (SFO–DEN) takes off from San Francisco, the last of the 18 evening flights to depart its origin.
- **21:39:** KDEN reports heavy thunderstorm rain, lightning, and visibility reduced to 1.75 miles.
- **21:53:** A routine KDEN report records heavy thunderstorm rain and hail, with visibility of 2 miles.

A Late-Night Wave of Diverted Flights at SLC

All times are MDT. The June 21 diversion count uses BTS `FlightDate`*, not the SLC landing date.*

- **22:09:** WN 955 (PDX–DEN) is the first of the 18 evening flights to land at SLC. It takes off again at **00:28** and reaches the DEN gate at **01:26** on June 22.
- **22:27** / **22:37:** WN 1308 and UA 1216 (both SFO–DEN) land at SLC ten minutes apart. They reach the DEN gate at **04:01** and **15:54** on June 22, respectively.
- **23:08: 16 of 18** evening flights have landed at SLC. DL 857 (DTW–DEN) lands at **23:34** on June 21; DL 956 (ATL–DEN) lands at **00:17** on June 22.
- **Subsequent outcomes:** Nine flights reach DEN overnight and eight later on June 22. For WN 2644 (SAN–DEN), the BTS record shows no subsequent SLC departure or DEN arrival.

Future Work: `slcweather`

- Develop `slcweather` as a companion R package to `slcflights`, beginning with the existing research scripts at https://github.com/AlKulAK/rgovys_talk_2026/tree/main/scripts.
- Provide reusable functions for GHCNh acquisition, historical airport–weather-station assignments, weather processing, and flight–weather linkage.
- Make time-aligned weather measurements, observation times, and original weather-report remarks available alongside reconstructed flight events.
- Extend the analysis to other diversion episodes, delays, and cancellations across SLC's flight network, using weather measurements as statistical covariates.

Future Work: `tidyflights`

- Make BTS data wrangling a matter of a few R commands: select airports and a study period, then build and query a local database.
- Automate field selection, downloads, ZIP extraction, and preparation. Hellwig (2014) documented the difficulties of obtaining BTS data manually.
- Complement flight-level On-Time Performance records with:
  - **T-100:** monthly passenger and freight totals by market; departures and capacity by nonstop segment.
  - **B-43:** annual carrier aircraft inventories, including models, seats, and tail numbers.
  - **DB1B/DB1C:** sampled passenger itineraries, fares, and ticket, market, and coupon records.
- Support research relating flight operations, traffic volumes, aircraft, and passenger journeys.

Collaboration and Research Opportunities

I welcome ideas and suggestions concerning this research, including possible extensions and additional applications.

I am also open to discussing potential collaborations and relevant funding opportunities.

If interested, please feel free to contact me or my PhD advisor:

- a02291810@usu.edu (Aleksey Kulyukin)
- juergen.symanzik@usu.edu (Jürgen Symanzik, PhD)

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Additional references appear in the appendix. The complete slidedeck is available in the GitHub repository: https://github.com/AlKulAK/rgovys_talk_2026.

Questions?



Image by geralt from Pixabay

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